

3. Solution Overview

CertChain works in two simple steps: the issuer registers a document hash on-chain; the verifier checks the hash. No intermediary, no phone call, no waiting.

Core principle: CertChain never stores the document itself on-chain. It stores only a SHA-256 hash of the document. The issuer retains the original. Any third party can verify authenticity in under three seconds.

3.1 Issuance Flow

- Authorised issuer (university, employer, government body) uploads the document or fills a structured template in the CertChain dashboard.
- The platform computes a SHA-256 hash of the document content + metadata (recipient name, issue date, expiry if applicable, issuer ID).
- A CertificateRegistry smart contract stores the hash on Polygon PoS mainnet with the issuer's wallet signature and a timestamp.
- CertChain generates a unique Certificate ID (UUID mapped to on-chain token) and a shareable verification link / QR code.
- The recipient receives a tamper-proof PDF or digital badge with the Certificate ID and QR embedded.

3.2 Verification Flow

- Verifier (recruiter, regulator, court, bank) receives the document and Certificate ID — or simply scans the QR code.
- CertChain's public verification portal computes the hash of the presented document and queries the smart contract.
- Result returned in under 3 seconds: VALID (hash matches, issuer confirmed) or INVALID (tampered / not found).
- No account, no login, no fee required for verification. Fully public.

4. Technical Architecture

4.1 Smart Contract Layer (Solidity on Polygon)

- CertificateRegistry.sol — core contract. Functions: issueCertificate(bytes32 hash, string metadata), revokeCertificate(bytes32 certId), verifyCertificate(bytes32 certId) returns (bool, address issuer, uint256 timestamp).
- IssuerRegistry.sol — whitelist of authorised issuers. Only admin-approved wallets can call issueCertificate. Prevents impersonation.